

QUBO Formulation: Binary SVM Training Toy Problem

Binary SVM Training Toy Problem

Problem Statement. Given a finite set of N candidate models, each associated with a precomputed classification loss ℓ_j and a regularization penalty r_j for $j \in \{0, \dots, N-1\}$, the task is to select exactly one candidate that minimizes the total regularized score. The objective is to find the candidate index j^* such that $\ell_{j^*} + r_{j^*}$ is minimized.

Decision Variables. Introduce binary selection variables $x_j \in \{0, 1\}$ for each candidate $j \in \{0, \dots, N-1\}$. The variable $x_j = 1$ indicates that candidate j is selected, and $x_j = 0$ otherwise.

QUBO Derivation. Step 1 — The original objective function in general notation is:

$$\min_x \sum_{j=0}^{N-1} (\ell_j + r_j) x_j. \quad (1)$$

Step 2 — The selection constraint requires that exactly one candidate is chosen:

$$\sum_{j=0}^{N-1} x_j = 1. \quad (2)$$

Step 3 — We convert the constraint to a quadratic penalty term with weight $\lambda > 0$. (a) The squared penalty is written as:

$$P(x) = \lambda \left(\sum_{j=0}^{N-1} x_j - 1 \right)^2. \quad (3)$$

(b) Expanding the square yields:

$$P(x) = \lambda \left(\sum_{j=0}^{N-1} x_j^2 + \sum_{j \neq k} x_j x_k - 2 \sum_{j=0}^{N-1} x_j + 1 \right). \quad (4)$$

Using the idempotent property $x_j^2 = x_j$ for binary variables, we collect terms:

$$P(x) = \lambda \left(\sum_{j=0}^{N-1} x_j + \sum_{j \neq k} x_j x_k - 2 \sum_{j=0}^{N-1} x_j + 1 \right) = \lambda \left(- \sum_{j=0}^{N-1} x_j + \sum_{j \neq k} x_j x_k + 1 \right). \quad (5)$$

(c) The resulting general penalty contribution to the Hamiltonian is therefore $\lambda \left(- \sum_{j=0}^{N-1} x_j + \sum_{j \neq k} x_j x_k + 1 \right)$.

Step 4 — Combining the objective and the penalty term gives the single QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{j=0}^{N-1} (\ell_j + r_j) x_j + \lambda \left(- \sum_{j=0}^{N-1} x_j + \sum_{j \neq k} x_j x_k + 1 \right). \quad (6)$$

Step 5 — Reading off the QUBO parameters from $H(x)$ in general closed form:

$$Q_{j,j} = \ell_j + r_j - \lambda, \quad \text{for } j \in \{0, \dots, N-1\}, \quad (7)$$

$$Q_{j,k} = 2\lambda, \quad \text{for } j, k \in \{0, \dots, N-1\}, j \neq k, \quad (8)$$

$$c = \lambda. \quad (9)$$

Penalty Weight Justification. Let $\Lambda = \max_{j \in \{0, \dots, N-1\}} (\ell_j + r_j)$. We set $\lambda = \Lambda + 1$. The maximum possible value of the original objective function over any feasible assignment is Λ . If the constraint is violated, the term $\left(\sum_{j=0}^{N-1} x_j - 1 \right)^2$ evaluates to at least 1, incurring a penalty of at least λ . Since $\lambda > \Lambda$, the energy cost of any infeasible configuration strictly exceeds the maximum possible objective gain achievable by any feasible configuration. Therefore, the global minimum of $H(x)$ cannot occur at an infeasible point.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^N$ recovers the optimal solution because the penalty weight λ is chosen to strictly dominate any feasible objective improvement. For any feasible assignment (where exactly one $x_j = 1$), the penalty term vanishes, and $H(x)$ reduces exactly to the original objective function. For any infeasible assignment, the penalty term contributes at least λ , which exceeds the maximum possible value of the original objective. Consequently, the global minimum of $H(x)$ must occur at a feasible point, and among feasible points, $H(x)$ is identical to the original objective, guaranteeing that the minimizer corresponds to the candidate with the lowest regularized score.

Illustrative Example. Consider an instance with $N = 2$ candidates. Let the precomputed scores be $\ell_0 + r_0 = 0.5$ and $\ell_1 + r_1 = 0.2$. The maximum score is $\Lambda = 0.5$, so we choose $\lambda = 0.5 + 1 = 1.5$. Substituting these values into the general QUBO parameters yields:

$$Q_{0,0} = 0.5 - 1.5 = -1.0, \quad (10)$$

$$Q_{1,1} = 0.2 - 1.5 = -1.3, \quad (11)$$

$$Q_{0,1} = Q_{1,0} = 2(1.5) = 3.0, \quad (12)$$

$$c = 1.5. \quad (13)$$

The resulting Hamiltonian is $H(x) = -x_0 - 1.3x_1 + 3x_0x_1 + 1.5$. Evaluating $H(x)$ over all $2^2 = 4$ binary assignments gives:

$$H(0, 0) = 1.5, \tag{14}$$

$$H(1, 0) = -1.0 + 1.5 = 0.5, \tag{15}$$

$$H(0, 1) = -1.3 + 1.5 = 0.2, \tag{16}$$

$$H(1, 1) = -2.3 + 3.0 + 1.5 = 2.2. \tag{17}$$

The global minimum is 0.2 at $x = (0, 1)$, correctly selecting candidate 1 with the lowest regularized score.